Paraphrase Generation and Deep Learning Models for Paraphrase Detection in a Low-Resourced Language: Kannada

H. M. Anagha, Karthik Sairam, Janya Mahesh, and H. R. Mamatha

1 Introduction

Paraphrasing is a natural language processing algorithm that replaces a given sentence with a sentence that communicates the same message as the original sentence. It is done with various purposes including better comprehension or correct representation of the content, depending on what one strives for while performing this reconstruction. Numerous opportunities can be incorporated into an artificial intelligence (AI) approach to provide a more interactive and effective communications platform. This is applicable in technical fields like medical science, engineering, and other sciences.

Understanding or interpreting the meaning of words is part of learning a language, and so is learning the rules governing sentences and grammar structures. However, paraphrasing is another chance to express once more the same thought but with other words and sentences that could result in better knowledge of how to properly use it.

Paraphrase Generation is an approach that generates sentences, which are grammatically and meaning-wise close to a selected exemplar of the same wording. One can extend this similar concept to a whole document with hundreds of lines or even thousands of them. Paraphrasing is one of those topics that have been enormously studied in depth concerning languages like English, French and Spanish. Such a system ought to exist. It should however set a benchmark so that other languages may come on board.

This indicates that there is a need for developing NLP techniques which will help solve NLP issues of low-resourced languages. This provides the basis for research in

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the Kannada language and over three million Kannada speakers will have an opportunity to interact with technologies which comprehend and talk back in their language. The ambitions of the authors include protecting and enhancing the Kannada dialect which will certainly be enhanced by the usage of a digital approach in promoting it.

There are two major segments when considering the implementation of the systems outlined under the title including; the paraphrase generation system and thereafter the deep learning for the paraphrase detection system. Conversion of text from the former is made possible through a T5 model, which converts an input text into output text. Due to the unavailability of appropriate datasets for such a task, the authors of this paper developed it themselves as the dataset. The operation of this process highly incorporates the tokenization of the data and their encryption as well as other attention self mechanisms. This work is an extension of the previous work of the authors of this paper, who were engaged in the creation of a paraphrase detection system in Kannada. This paper proposes a new deep learning technique that can be used to enhance the learning speeds and accuracies of models for improved outcomes. Therefore, a multilayer neural net was used. In addition, a number of optimizers was also employed side by side so as to make the data fit into the model.

2 Literature Survey

The authors of [1] articulate the architecture for producing a paraphrases framework. This is where they incorporate the deep generative model, plus the seq2seq model for the best results from the Quora Question dataset and MSCOCO (Microsoft Common Objects in Context) dataset. They also deliberate on the pitfalls of automatic paraphrase production. Therefore, the author proposes using the VAEs (Variational Autoencoders), which resemble the encoder/decoder side of the T5 model employed by this paper's authors, to generate several principle paraphrases. In addition, the developed paraphrases are not only semantically equivalent to the input sentence but are capable of expressing additional ideas associated with the initial sentence. In the course of providing multiple paraphrases for an input sentence, the weakness of the model becomes evident in that it tries to expand the scope of the information. Deviation in the literal sense would also be possible which could then alter the meaning of the paraphrased sentence and the actual motive behind the input one.

The author of [2] utilizes the same model (the T5 model) for two tasks—paraphrasing and detecting paraphrasing. Finally, they use the encoder-decoder T5 model that tunes its hyperparameters in the end to generate the text output. This is achieved on ParaNMT, Msrp, and Quora duplicate question pairs datasets. This is somewhat of a high-level research, which has superior performance compared to analogous systems in reviews. This is done in English. Further, as the model outputs various paraphrased sentences, a particular sentence from the output may be identical with the source sentence. The reliability of this type of model thereby definitely decreases due to that reason.

The authors of [3] are aimed at creating a holistic paraphrasing system for the Hindi language. For this, they have developed recurrent neural networks-based attention models. The system additionally utilizes LSTM, adaptive attention models, and GRUs. Among the methods invented are synonym substitution and transformation of the active into the passive voice. This is why both LSTM and GRU were used as they solve the problem of exploding and vanishing gradients of RNN. However, this kind of system can be weak only because of its generality.

In the feature extraction phase, the authors used an N-gram-based paraphrase generator described in [4]. The machine learning model was excellent due to numerous Sanskrit language conforming rules. The algorithm could break a compound word into two simple words, determine the meaning of the compound word, and then choose either to retain the compound or substitute it with its paraphrase. Model was able to read and interpret fine details of Sanskrit which translated highly efficient output. One of the main positive sides of the ML model is that it can describe the intricacy of the Sanskrit language in detail. It can consider many rules of grammar and syntax which determine the structure within a language enabling it to interpret and translate the text effectively. Its other strength is the application of methods for identifying the gender, numeral, and grammatical ending of the final part of complex nouns. The depth and sense involved during this process enables the model to generate faithful, precise and effective translations of the Sanskrit language making it a dependable model during operations involving this language. There are a few issues with using the machine learning model on tasks related to the Sanskrit language that should be considered. An example of this drawback is, that the model will at times separate the different parts of the same word depending on its gender. This could result in mistakes in a transliteration or even have an increased correction demand in a manual. Also, there might be a possibility that the model will depend on the completeness of its dictionary and if the morphological analyzer fails to provide true analysis. In other words, these limitations can sometimes lead to partial or faulty translations; however, the system itself remains to be useful for various tasks dealing with this particular kind of language.

The aim of the authors of [5] is to create a dataset in the Bangla language, that helps to further the research in the Bangla language and helps in promoting NLP tasks such as machine translation and text classification. This is achieved through an extremely diverse dataset built in the Bangla language. This self-made dataset was scraped from a Bangla web domain and was processed by annotating the sentences that are paraphrases (especially those with the same meaning but different words)

In addition, they also built 4 novel filtering systems that are pipelined, based on the different parameters that are required to analyze a sentence. Hence, it can be said that each filter might cover the downfalls of the other filters, and can thus summarize the entire filtering system efficiently and appropriately.

This paper by Fu et al. [6] presents a new way to generate paraphrase using LDA generative model. The authors present paraphrase generation as a learning process of a probabilistic mapping from sources to targets, introducing a generative model that restores sources through production of paraphrases. Based on LDA (widely used model for unsupervised topic modeling) with bag-of-words representation of the given sentences learning in latent space. They demonstrate an efficient and high standard generation model whose quality and number are better than those presented by today's leading technology methods.

This paper presents a new approach for paraphrase development which integrates unsupervised topic modeling, and a bag-of-words model based generative technique. The authors demonstrate that this method can produce highly informative and multi-variant paraphrasing hence a powerful instrument in NLP processes including text generation, question answering or text categorisation. The writers demonstrate that the suggested approach could be used in various languages and domains. This develops a more universal way of creating paraphrases and paves way for further research into this topic.

The paper [7] reviews existing metrics for scoring quality in generated sentences. This makes it difficult for the authors of this paper to argue that paraphrasing generation system evaluate is hard. At last, these findings help us appreciate the shortcomings associated with the existing evaluation criteria used in measuring success of software projects.

The current state of paraphrase generation and its valuation techniques are summarized in this broad review. The article details the strengths and shortcomings of those metrics as well. The metrics in this paper are classified into three main categories:

- Intrinsic metrics depend upon the properties of the created paraphrases and extrinsic measures that gauge effects of produced alterations on downstream NLP tasks.
- Secondly, there are the human evaluations, which rely on human judgments.

The authors finally take an in-depth look at those metrics and demonstrate that there is a tradeoff between feasibility and accuracy in which case determines the most appropriate metric for assessment of paraphrase generation systems.

Using the MSRP corpus on a dataset, this paper [8] used 14 stringed similarity measures and four corpus based similarity measures. These constitute properties that are inputs to the plagiarism detector system. The implementation of test and train functions in the paraphrase system comes using Weka tool, having 50 classifiers. It also proposes two hybrid methods of combination namely, combinetall and combinetbest. To analyze how various measures take in sentences, we compare the two different models.

The paper [9] provides a basic understanding of paraphrase generation. It talks about the importance of a good dataset, evaluation methods, traditional approaches (without any neural models), and some neural approaches used for paraphrase generation. In addition to this, it also provides a comparison between all the paraphrase generation methods.

The datasets used for paraphrase generation are PPDB, Quora, MSCOCO, WikiAnswer, Twitter URL, and ParaNMT. Human evaluators or automated evaluators can be used to evaluate the paraphrase generator. Some Traditional approaches to conducting the same are Rule-Based, Thesaurus-Based, and SMT-Based. Neural approaches include an encoder-decoder architecture. Usually, the encoder extracts semantic features which are then passed on to the decoder which contextually represents it. Improvements to this architecture are made using Variational autoencoders, reinforcement learning, and Generative adversarial networks. In conclusion, this paper gives an excellent overview of paraphrase generation.

The authors of the paper [10] propose a novel approach for paraphrase generation. PPDB, WikiAnswers and MSCOCO are the three datasets used in their paper. The novel neural approach consists of stacking LSTM models. Although this gives rise to a degradation problem, residual connections solve this issue. SGD was used to train the models and BLEU, METEOR, and TER were used as evaluation metrics. The best results were obtained while using a residual LSTM with 4 layers.

The paper [11] proposes a novel framework TranSeq, that combines transformer and seq2seq model for paraphrase generation. This model has been evaluated on the Quora and the MSCOCO dataset. TranSeq has the efficiency of a transformer and seq2seq improves the results. TranSeq uses the encoder-decoder architecture. It has a stacked encoding layer consisting of transformer and GRU-RNN. The paper also provides a comparison between many state-of-the-art models for paraphrase generation and TranSeq. The evaluation metrics used are BLEU, METEOR and ROUGE. TranSeq outperforms all the other models.

3 Proposed Methodology

Both the tasks of Paraphrase Generation and Deep Learning implementation for Paraphrase Detection are implemented using Python 3.x and various other Python libraries like INLTK (Indic Natural Language Toolkit), T5, and Pytorch. In addition, the Hugging Face framework is used to implement the end-to-end paraphrase generator.

3.1 Dataset Creation and Acquisition

The dataset used for both tasks consists of two parts, the dataset that was manually crafted by the authors, and the MSRP (Microsoft Research Paraphrase) corpus.

The created dataset consists of 1600 data points, in the Kannada language. It contains news headlines and parables, some of which are paraphrases and some of which are not paraphrases. The indicator of whether a data point has two paraphrases or not is given by its ’Quality’, a column in the dataset. These values were manually entered by the authors and also validated for maximum readability and appropriate grammar pertaining to the sentences.

The MSRP corpus has 5800 data points and is originally in English, hence a translator automation tool had to be built so as to translate them into the Kannada language. Following this, the data points have to be validated for all the sentences whose meanings were lost above a certain threshold value during translation. This was also carried out manually by the authors for two reasons,

Sentence_1	Sentence_2	Quality
ಪಿಸಿಸಿಡಬ್ಲ್ಯೂನ ಮುಖ್ಯ ಕಾರ್ಯನಿ	ಪ್ರಸ್ತುತ ಮುಖ್ಯ ಕಾರ್ಯನಿರ್ವಹಣಾ	1
ಬೇಸಿಗೆಯ ಕೊನೆಯಲ್ಲಿ ಮಾರಾಟದ	ಮತ್ತು ನಂ ಫೋರ್ಡ್ ಮೋಟಾರ್ ಕ	1
ಫೆಡರಲ್ ಸೆಂಟರ್ಸ್ ಫಾರ್ ಡಿಸೀಸ್	ರಲ್ಲಿ ಯುನೈಟೆಡ್ ಸ್ಟೇಟ್ಸ್‌ನಲ್ಲಿ ದಡಾ	1
ಗಲ್ಫ್ ಆಫ್ ಮೆಕ್ಸಿಕೋದಲ್ಲಿ ಉಷ್ಣವ	ಭಾನುವಾರ ಗಲ್ಫ್ ಆಫ್ ಮೆಕ್ಸಿಕೋದ	0
ಕಂಪನಿಯು ಬದಲಿ ಮತ್ತು ರಿಪೇರಿ	ಆದರೆ ಕಂಪನಿಯ ಅಧಿಕಾರಿಗಳು ಬ	0
ಇತ್ಯರ್ಥಪಡಿಸುವ ಕಂಪನಿಗಳು ಹೂ	ಒಪ್ಪಂದದ ಅಡಿಯಲ್ಲಿ ಇತ್ಯರ್ಥಪಡಿ	1
ಏರ್ ಕಮೋಡೋರ್ ಕ್ಲೈಫ್ ಅವರು	ಭದ್ರತಾ ಕಾರ್ಯಾಚರಣೆಯು ಅಭೂ	0
ವಾಷಿಂಗ್ಟನ್ ಕೌಂಟಿಯ ವ್ಯಕ್ತಿಯೊಬ	ಈ ವರ್ಷ ವೆಸ್ಟ್ ನೈಲ್‌ನ ಕೌಂಟಿಗಳ	1
ಮೊಸ್ಲಿ ಮತ್ತು ಹಿರಿಯ ಸಹಾಯಕರು	ನೆವಾಡಾದ ನೆಲ್ಲಿಸ್ ಏರ್ ಫೋರ್ಸ್	1
ವಿಶಾಲವಾದ ಸ್ಟ್ಯಾಂಡರ್ಡ್ & ಪೂ	ಟೆಕ್ನಾಲಜಿಲೇಸ್‌ನ ನಾಸ್ಡಾಕ್ ಕಾಂಪೋ	0

Fig. 1 Dataset created

- The lack of availability of tools that can validate the structure of a Kannada sentence
- To be precise and thorough in validation to prevent inconsistencies during the later stages of the implementation.

The final dataset, consisting of both the above parts, has a structure that is shown in Fig. 1. The final dataset also has a total of over 7400 data points.

3.2 Paraphrase Generation

The task of paraphrase generation is implemented using the multilingual T5 model (Text-To-Text Transfer Transformer) [12]. This has been trained in 101 languages. This model can be used to do machine translation on the created Kannada corpora. mT5 model follows the encoder-decoder architecture. It can be trained in a supervised or unsupervised manner. For training an mT5 model, we need an input sentence and a corresponding output sentence. Figure 2 is a block diagram of the approach used for paraphrase generation.

From the corpora created, the sentences which were paraphrased (i.e. had Quality = 1) were selected. These were given as the input to the mT5 model. Column Sentence_1 was given as the input sentence and Sentence_2 was given as the corresponding output/target sentence to the mT5 model.

This task is implemented using the T5 model that does the work of machine translation on the Kannada dataset. Fine-tuning the T5 model helps customize the parameters to the use case, which is paraphrase generation. To accomplish this task, a PyTorch Lightning training script, which defines, and imports key functions and libraries is run. The main class has various methods for performing training, validation, and testing of the model. Further, it also has a forward method which computes the model's forward pass on the given input. The above task is CPU-intensive, hence, seeds were introduced in the code to ensure that the execution occurs without any memory shortages and runtime errors.

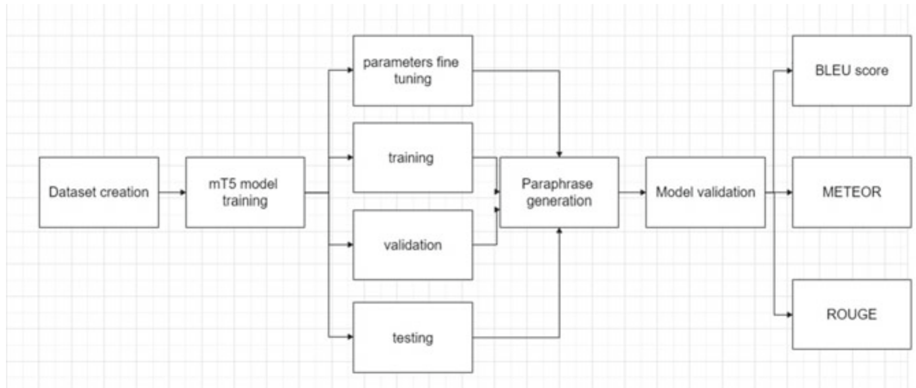

Fig. 2 Pictorial representation of the approach

The architecture of such a system involves various components, including:

- A feed-forward network
- Self-attention mechanism at every layer of the model
- Pre-processing component
- Post-processing component.

The inputs given are processed by the self-attention mechanism. The feed-forward network then generates outputs corresponding to the processed inputs. The attention mechanism plays a vital role in providing a better understanding of the context behind each sentence input in Kannada. It also tags dependencies between words in the input sentence so that the closest possible meaning can be taken.

The most important step also happens to be the first step, which is fine-tuning the parameters. The dataset explained in the previous section is used as the input for fine-tuning the model. This step ensures that the model learns the patterns and relationships between the tokens. Since this task has a lot of processing, it is a GPU-intensive task and requires multicore computing.

Hyperparameter tuning is the next crucial step. This is done to optimize the functioning of the model. This step includes making small changes to the hyperparameters(batch size and learning rate). After the completion of this step, the model is ready to be evaluated based on its performance for a wide variety of diverse Kannada sentences. Evaluation metrics like BLEU score have been used for the same purpose.

3.3 Paraphrase Detection Using Neural Networks

Paraphrase detection is an NLP task that checks whether 2 sentences are synonymous. The goal is to implement this in the official Kannada language using Neural networks.

Features need to be extracted from the corpora created. These features are then passed as inputs to the neural networks. The features extracted from the corpora are:

Table 1 Accuracy achieved using various optimizers

Optimizers	Accuracy
Adam	0.7034
SGD	0.6763
Adagrad	0.6443

- Cosine Similarity
- Jaccard's Similarity
- Cosine Similarity with Bag Of Words
- Levenshtein's distance
- Jaro-Winkler
- Needleman-Wunsch.

A Simple Neural network was created for this classification problem. The neural network is of sequential type and it consists of 2 dense layers with ReLU as its activation function. The last dense layer is the output layer which uses Softmax as its activation function. The optimizer used was Adam. The learning rate was 0.001. Categorical cross entropy was the loss function. The model had a total of 202 trainable parameters. The accuracy of the model was 0.7034. The same was done for different optimizers such as SGD and Adagrad. The results of all the accuracies can be seen in Table 1. The best-performing optimizer is Adam. Since Adam has the fastest convergence when compared to SGD and Adagrad, it is best suited for our problem.

4 Results and Discussion

The output of the model is a paraphrase of the input sentence seen in Fig. 3.

Evaluation of a model is an extremely necessary step. Evaluation metrics are used to evaluate a model. The evaluation metrics chosen for this model are:

Fig. 3 Sentences generated from the paraphrase generator

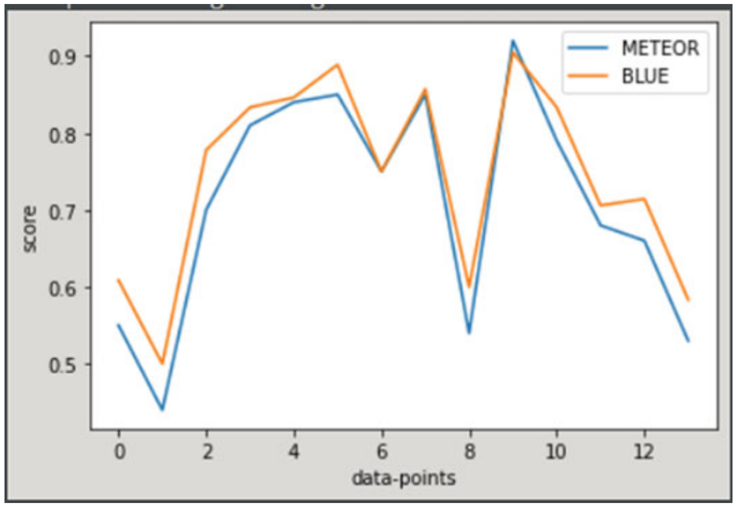

Fig. 4 Comparison of BLEU and METEOR scores

Table 2 Evaluation metrics rouge score

	Recall	Precision	F1-Score
ROUGE-1	0.7043	0.7411	0.7195
ROUGE-2	0.5012	0.5327	0.5137
ROUGE-L	0.6610	0.6958	0.6752

- BLEU (BiLingual Evaluation Understudy)
- METEOR (Metric for Evaluation of Translation with Explicit ORdering)
- ROUGE (Recall-Oriented Understudy for Gisting Evaluation).

The BLEU score of the model is 0.7078 and the METEOR score obtained is 0.7431. Figure 4 gives us a visual representation of how the model performs against BLEU and METEOR. Table 2 summarizes precision, recall and f1-score obtained by rouge-1, rouge-2, and rouge-L evaluation metrics. The comparison between the recalls obtained from the ROUGE evaluation metric can be seen in Fig. 5.

5 Conclusion

It becomes quite difficult if one wants to generate paraphrases in a low-resourced language like Kannada where tools and datasets are not easily accessible. As such, this article tries to accomplish this goal with the aid of the mT5 model. With respect to the mt5 model, the authors obtained a METEOR score of 0.7431 and a BLEU score of 0.7078. However, the evaluation metrics have become stable for this task of paraphrase generation such that it is an integral element of future NLP applications

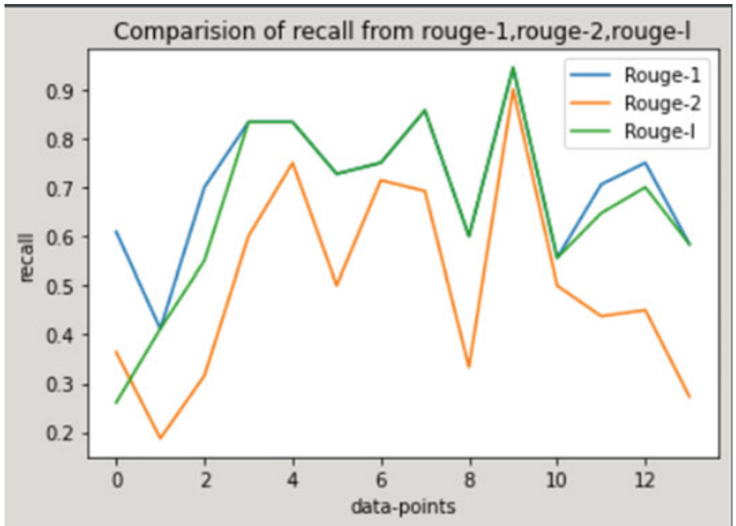

Fig. 5 Comparison of recall from rouge-1, rouge-2, rouge-l

in the Kannada language. It has the ability to break down complicated notions, help one understand words better, translate from one language into another, and give machines good translation for processing. Therefore, there should be more research and development done on the Kannada language paraphrasing techniques that can boost the improvement and usage of this language.

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